

Quantifying Modality Inertia in Medical Vision-Language Models: Image Contribution Metric, Gate Stagnation, and the Failure of Debiasing Under Label Circularity

Nafis Al Rahman, Tanvir Mahmud

ARTICLE INFO

Keywords:

medical vision-language model
modality inertia
gate stagnation
Image Contribution Metric
label circularity
MIMIC-CXR

ABSTRACT

Background. Medical vision-language models can report high accuracy while relying on report text rather than radiographic evidence. This risk is especially difficult to interpret in MIMIC-CXR, where CheXpert targets are generated from the same reports that may remain available as model inputs. We evaluated whether text-debiasing increases incremental image information or merely changes aggregate performance.

Methods. We audited a three-class MIMIC-CXR-derived classifier comprising Normal, Pneumonia, and Pneumothorax across eight random seeds and five conditions: baseline gated fusion, a learned text-prior product-of-experts variant, a noise-sensitivity variant, an exploratory grounding-aware variant, and an adaptive-gate variant. Each condition used 455 held-out studies; report-preserving Gray-Square replacement, text-only, and image-only controls were retained. The primary endpoint was the Image Contribution Metric (ICM), with a pre-specified equivalence interval of minus 0.01 to plus 0.01. We combined paired seed-level inference with calibration, gate extraction, and qualitative saliency checks.

Findings. Baseline Macro-F1 was 0.9397 plus or minus 0.0061, compared with 0.9381 for the retained Gray-Square control. Baseline ICM was 0.0016 plus or minus 0.0065; the signed-rank test was non-significant ($p = 0.4219$), while two one-sided tests supported equivalence to zero within the operational interval ($p = 0.0040$). No text-debiasing condition demonstrated a robust increase in ICM. The text-prior variant increased expected calibration error from 0.0237 to 0.0621 and Brier score from 0.0522 to 0.0685. Extracted baseline gates remained near initialization, with an overall mean of 0.49975.

Interpretation. On this report-linked benchmark, near-ceiling performance was compatible with a report shortcut and did not establish incremental image grounding. The result is a diagnostic warning about label-report circularity and endpoint design, not evidence that radiographs are universally unnecessary. Independent labels, corrected implementations, matched controls, provenance manifests, and external cohorts are required before making broader clinical claims.

1. Introduction

Medical vision-language models (VLMs) combine radiographs with the clinical reports that accompany them. This pairing is valuable because images encode visual structure whereas reports provide findings and context; resources such as MIMIC-CXR make both modalities available at scale (Johnson et al., 2019). High multimodal accuracy, however, does not establish that the model used clinically meaningful visual evidence. A model may exploit a report shortcut or another predictive feature that is easier to optimise than image information.

This distinction is especially important when targets are report-derived. The CheXpert labeler converts report language into uncertainty-aware chest-radiograph labels (Irvin et al., 2019). If the same report, or its findings field, is also supplied to the model, the target and one input modality share an information source. The resulting task may reward linguistic recognition of the report-derived label rather than independent image-grounded recognition. Report-conditioned prediction can still be useful; the problem is claiming incremental image evidence from aggregate accuracy alone.

Recent evidence shows why this distinction matters in medical VLM evaluation. Performance can change with the amount of clinical context supplied before an image is seen, and apparently correct multimodal answers may be accompanied by unreliable visual rationales (Wang et al., 2026; Jin et al., 2024). Large-scale chest-radiography audits have likewise found that removing the image can leave performance largely intact, while consistency across prompts can coexist with non-visual decision making (Lotfinia et al., 2026; Sadanandan and Behzadan, 2026). Counterfactual prototype methods can improve interpretability, but they do not replace a report-preserving intervention that directly measures the change in the prediction (Chiang, 2026).

Conventional model comparisons rarely replace the image while holding the report, label, architecture, and evaluation protocol fixed. Similarly, a non-zero fusion gate or attention map is not necessarily causal evidence of visual use: gates can remain near initialization and a one-token interaction cannot show word–region alignment. Text-debiasing methods may therefore lower a score without creating an independent visual signal when the endpoint itself is linked to text.

We present a controlled audit of a three-class MIMIC-CXR-derived task (Normal, Pneumonia, and Pneumothorax). The model receives a frontal radiograph and the report’s findings field. We compare full multimodal models with a report-preserving Gray-Square control that replaces every radiograph by a constant tensor, together with text-only and image-only controls. Four fusion or text-debiasing variants are evaluated alongside the project’s baseline gated dual-path model (CADQ). We report discrimination, calibration, class-specific behaviour, learned image gates, and qualitative gradient-weighted class activation mapping (Grad-CAM) checks. The primary diagnostic is the Image Contribution Metric (ICM), defined as the relative Macro-F1 difference between the full model and the Gray-Square reference; paired inference is performed at the trained-seed level because held-out studies recur across seeds.

The audit asks whether removing image information changes performance, whether the interventions increase incremental image contribution, whether checkpoint gates show learned reliance or stagnation, and whether calibration and class-specific results qualify the headline metrics. It contributes a report-preserving image ablation, seed-level uncertainty and calibration reporting, and an explicit analysis of label provenance and control asymmetry. The study is a diagnostic evaluation, not a new clinical predictor or a claim that radiographs are generally unnecessary. Its interpretation is limited to this report-linked cohort, preprocessing path, and checkpoint archive; broader claims require corrected implementations, independent labels, matched controls, and external validation. Reporting follows the transparency principles of the Checklist for Artificial Intelligence in Medical Imaging (CLAIM) and the TRIPOD+AI framework (Mongan et al., 2024; Collins et al., 2024).

2. Related Work

2.1. Medical image–text learning and report-derived targets

Paired image–report resources support contrastive and local alignment methods such as ConVIRT and GLoRIA, while clinical-domain language models improve the handling of negation and uncertainty (Zhang et al., 2022; Huang et al., 2021; Boecking et al., 2022). These methods establish the value of reports but do not show that a downstream prediction depends on the image. MIMIC-CXR links radiographs to free-text reports, whereas CheXpert derives uncertainty-aware labels from report language (Johnson et al., 2019; Irvin et al., 2019). Report-labeling tools can also propagate errors and subgroup differences (Sabottke et al., 2024; Santomartino et al., 2024). The present audit therefore distinguishes report-conditioned prediction from independent image-grounded diagnosis when both target and text input share a report source.

2.2. Shortcut learning and unimodal bias

Shortcut-learning research shows that high in-distribution accuracy can arise from predictive but unintended features (Geirhos et al., 2020). In radiology, acquisition devices, population structure, annotation practice, and clinical context can create such shortcuts (Banerjee et al., 2023). In visual question answering, VQA-CP, GVQA, product-of-experts methods, and RUBi reduce answer or language priors by encouraging visual evidence (Agrawal et al., 2018; Clark et al., 2019; Cadene et al., 2019). They motivate LMH-style debiasing, but do not resolve the harder case in which the benchmark label is itself text-derived.

2.3. Auditing modality reliance

Recent multimodal evaluations separate correctness from evidence dependence: correct answers can be accompanied by flawed visual rationales, and stable predictions need not be image-reliant (Jin et al., 2024; Sadanandan and Behzadan, 2026). Chest-X-ray studies further report heterogeneity after controlling for clinical context (Wang et al., 2026). Causal audits using image occlusion, cross-patient replacement, and text-only comparisons provide a direct precedent for testing image use (Lotfinia et al., 2026). Prototype and counterfactual methods can improve saliency, but heatmaps alone cannot establish that image removal changes a decision (Chiang, 2026).

2.4. Fusion gates and explanation limits

Gated multimodal units learn modality weights, yet a non-zero or balanced gate is not causal attribution (Arevalo et al., 2017). A gate may remain near its initialization when the objective does not identify a different value. Likewise,

reasoning traces or plausible radiologist-style explanations are useful interfaces but not grounding tests (Myronenko et al., 2025). The present study combines checkpoint gate extraction with a report-preserving image intervention and treats Grad-CAM as a qualitative sanity check.

2.5. Positioning of the present study

Prior work treats image–text learning, report-label uncertainty, shortcut bias, and behavioural audits separately. We examine their intersection using report-preserving, text-only, and image-only controls, an operational ICM, checkpoint gates, and seed-level inference. The study tests incremental visual evidence without claiming that report-conditioned prediction or all medical VLMs are invalid.

3. Methods

3.1. Study design and analysis scope

We conducted a controlled audit of a medical vision–language classifier. The estimand was the incremental image contribution when the report remained available. The principal intervention replaced each radiograph with a uniform gray tensor while retaining the text, label, architecture, and evaluation protocol. Full multimodal predictions were compared with this control and with text-only and image-only models.

The replication unit for confirmatory inference was a trained model at a fixed seed. Eight seeds (42, 123, 456, 789, 1010, 1111, 1212, 1313) were used for baseline CADQ, the LearnedMixinFusion text-prior variant (LMH), Noise-Consistency, and Adaptive-CADQ. The exploratory grounding-aware variant (GACR-B) was added after the initial plan. Because the same test studies recur across seeds, subjects were not treated as independent observations across seeds. ICM was the primary diagnostic; calibration and gate behaviour were secondary outcomes. The Adaptive-CADQ extension and GACR-B are labelled post hoc.

Figure 2 records the complete cohort-construction path and the patient-level analysis split. Figure 1 summarizes the corresponding audit workflow from data provenance through paired inference.

Methodology workflow: auditing incremental image evidence under label-report circularity

MIMIC-CXR three-class task | fixed report text | matched controls | seed-level inference

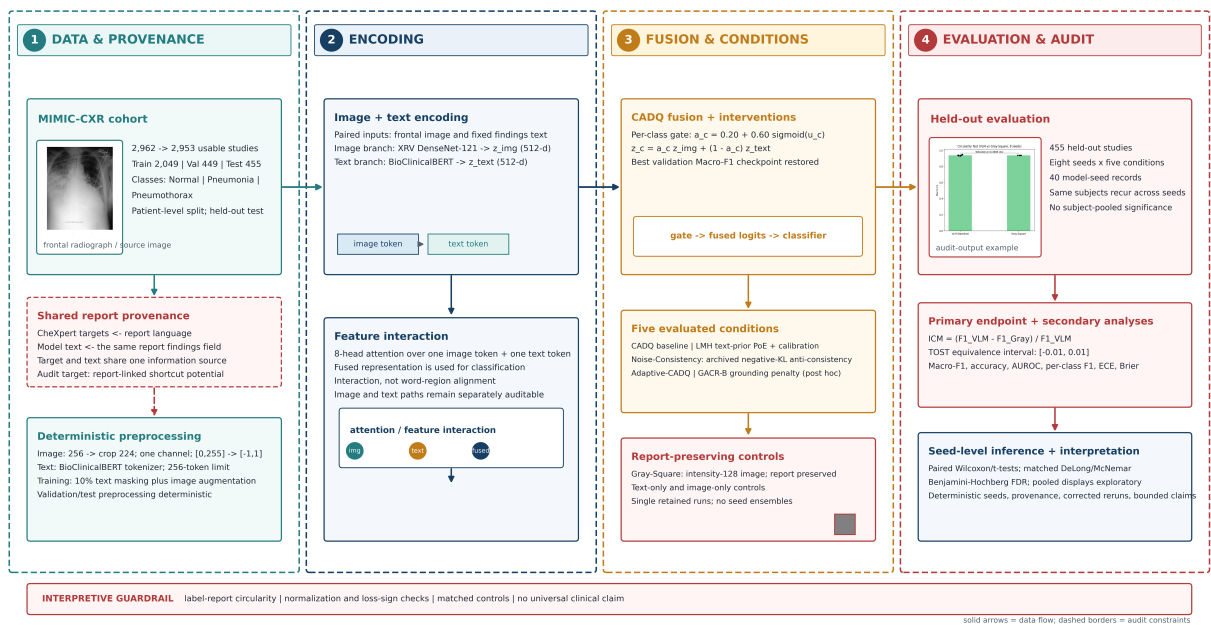


Figure 1: Audit workflow from MIMIC-CXR data provenance and patient-level splitting through image/text encoding, CADQ fusion, report-preserving controls, and seed-level paired inference.

Cohort construction and patient-level analysis split

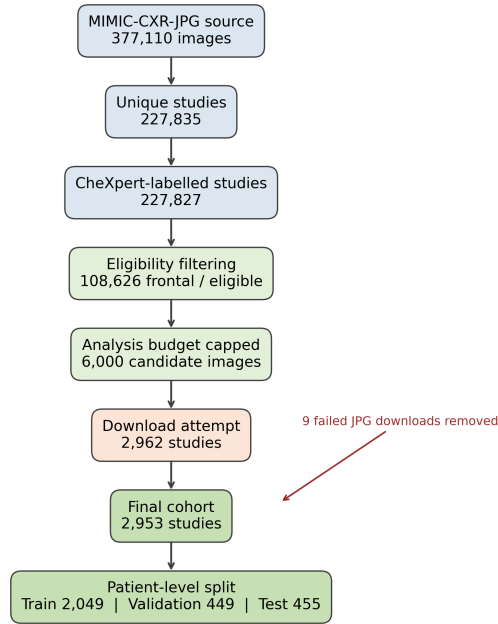


Figure 2: Cohort construction and patient-level analysis split. Nine failed JPG downloads were removed before the final 2,953-study cohort was partitioned.

3.2. Data source and cohort construction

MIMIC-CXR JPG images and reports were accessed through PhysioNet credentialing and the applicable Data Use Agreement (Johnson et al., 2019). From 2,962 candidate records, nine missing JPG files were removed, leaving 2,953 studies (2,049 training, 449 validation, and 455 test). Only frontal radiographs were retained. The three classes were Normal, Pneumonia, and Pneumothorax; class caps were 2,500, 2,500, and 1,000, respectively. The inherited patient-level split contained no subject overlap and 47 Pneumothorax studies in the test set. For transparency, the source-to-analysis flow was 377,110 JPG images, 227,835 unique studies, 227,827 CheXpert-labelled studies, 108,626 eligible frontal studies, a 6,000-candidate analysis cap, 2,962 download attempts, and 2,953 retained studies after the nine download failures (Fig. 2).

CheXpert generated the targets from report language using the project’s uncertain-to-negative mapping (Irvin et al., 2019). The model received the same report’s findings field. Thus, the experiment audits label–report circularity rather than an independent image-only diagnostic task. This secondary analysis used de-identified data; no participants were recruited and no direct identifiers were accessed. MIMIC-CXR was released for credentialed research under the source-study IRB approvals; no additional local participant recruitment or intervention was undertaken for this audit.

3.3. Preprocessing

Findings text was tokenized with the emilyalsentzer/Bio_ClinicalBERT tokenizer and truncated or padded to 256 tokens. During training, 10% of strings were replaced by an empty string; validation and test text was unmasked. Images were resized to 256×256 , randomly cropped to 224×224 , converted to one channel, and mapped from $[0, 255]$ to $[-1, 1]$. Training used horizontal flips, rotation (up to 10°), translation (5%), and brightness/contrast jitter (0.2); the class-aware minority transform increased rotation to 15° and used translation up to 10%, scale 0.9–1.1, shear up to 10° , stronger jitter (0.3), and Gaussian blur. Validation and test images were deterministically resized and mapped. The archived $[-1, 1]$ mapping may not match the native TorchXRayVision convention and requires a corrected rerun (Cohen et al., 2022).

3.4. Model architecture

The image branch used TorchXRayVision (XRV) DenseNet-121 with `densenet121-res224-mimic_ch` weights (Huang et al., 2017; Cohen et al., 2022). Its output was projected to 512 dimensions with linear, LayerNorm, Gaussian Error Linear Unit (GELU), and dropout (0.25); the convolutional stem and first dense block were frozen. The text branch used Bio_ClinicalBERT, with its first four layers frozen, and an analogous 512-dimensional projection (Devlin et al., 2019; Alsentzer et al., 2019). The projected image was one query token and projected text one key/value token in an eight-head batch-first attention block with residual LayerNorm. This is a feature interaction, not token-level word–region alignment; its single-key attention is degenerate and is not interpreted as causal grounding.

3.5. Fusion variants and controls

CADQ produced image and text logits and combined them with a learnable per-class gate (Arevalo et al., 2017):

$$\alpha_c = 0.20 + 0.60 \text{sigmoid}(u_c), \quad z_c = \alpha_c z_c^{\text{img}} + (1 - \alpha_c) z_c^{\text{text}}, \quad (1)$$

with $u_c = (-0.5, 0, 0.5)$ and initial gates approximately $(0.4265, 0.5000, 0.5735)$. LMH added a five-fold out-of-fold term-frequency/inverse-document-frequency (TF–IDF) logistic text-prior branch (5,000 unigram/bigram features, stop-word removal, balanced weights), validation temperature calibration, $\epsilon = 0.1$ smoothing, an input-dependent prior gate, and an entropy penalty (weight 0.05, annealed for three epochs) following the product-of-experts debiasing rationale of Cadene et al. (Cadene et al., 2019). It was warm-started from CADQ.

Noise-Consistency added Gaussian image noise ($\sigma = 0.25$) and the archived term $\mathcal{L}_{NC} = -D_{KL}(p_{\text{clean}} \| p_{\text{noisy}})$ based on Kullback–Leibler (KL) divergence to a minimised loss. It therefore rewards divergence and is reported as anti-consistency/noise sensitivity; a positive-KL rerun is required for a conventional consistency claim. Adaptive-CADQ replaced the static gate with a sample-dependent scalar from image/text entropies through a $2 \rightarrow 16 \rightarrow 1$ multilayer perceptron (MLP). GACR-B used an out-of-fold text baseline and a cosine penalty (weight 0.3) on examples where that baseline was wrong; it is exploratory and follows the grounding-aware consistency intervention of Sadanandan and Behzadan (Sadanandan and Behzadan, 2026).

All controls used the same test studies. Image-only used the XRV encoder and a 512-to-256 classifier; text-only used frozen Bio_ClinicalBERT and a 768-to-256 classifier. Gray-Square replaced each image with a constant intensity-128 tensor while preserving the report and multimodal path. Retained image-only and Gray-Square outputs were single runs, not seed ensembles.

3.6. Training and checkpoint selection

Models were trained for 10 epochs with AdamW (Loshchilov and Hutter, 2019), differential learning rates of 10^{-5} for image/text backbones, 10^{-4} for fusion and LMH heads, and 5×10^{-5} for static gates; weight decay was 10^{-4} . OneCycleLR used 10% warm-up (Smith and Topin, 2019). Physical batch size was 8 with four accumulation steps (effective 32), mixed precision, and gradient scaling. Training used focal cross-entropy ($\gamma = 1.5$), label smoothing 0.05, and effective-number weights ($\beta = 0.99$) (Lin et al., 2017); class-aware sampling replicated Pneumothorax up to fivefold and Pneumonia up to threefold. The best validation Macro-F1 checkpoint was restored for multimodal variants. The archived image-only control did not consistently restore its best state and is descriptive.

3.7. Endpoints and statistical analysis

The primary endpoint was

$$\text{ICM} = \frac{F1_{VLM} - F1_{Gray}}{F1_{VLM}}, \quad (2)$$

where positive values indicate an incremental gain from the real image. The two one-sided tests (TOST) procedure tested equivalence to zero within $[-0.01, 0.01]$, an operational margin rather than a clinical threshold (Lakens, 2017). Per seed, we computed macro-averaged F1 score (Macro-F1), per-class F1, accuracy, one-versus-rest area under the receiver operating characteristic curve (Macro-AUROC), confusion matrices, ten-bin expected calibration error (ECE), and multiclass Brier score. Paired Wilcoxon tests (Pratt zero handling), paired t -tests, bootstrap intervals, and Cohen’s d were used for matched seed differences; ECE/Brier were recomputed from retained probability arrays and absent outputs were recorded as N/A, never zero.

DeLong AUROC comparisons for correlated curves (DeLong et al., 1988) and McNemar tests for paired classification decisions (McNemar, 1947) were performed within matched seeds; studies were never concatenated

across seeds for confirmatory inference. Any pooled display is exploratory. Benjamini–Hochberg controlled false-discovery rate within each test family (Benjamini and Hochberg, 1995), with two-sided $p < 0.05$ nominally. Gate values were extracted from saved checkpoint logits using the bounded equation above; manual or hard-coded plot values were excluded.

3.8. Reproducibility safeguards

Python, NumPy, and PyTorch seeds were fixed and deterministic cuDNN settings enabled. Run metadata, validation history, checkpoints, predictions, probabilities, and configurations were saved. The fixed archive contains duplicate merge keys without a preserved source-priority rule or hash manifest; results are therefore tied to this snapshot and require provenance locking. The normalisation issue, negative-KL implementation, unequal post-hoc history, and image-only checkpoint discrepancy are reported explicitly. Reporting follows CLAIM and TRIPOD+AI (Mongan et al., 2024; Collins et al., 2024).

4. Results

4.1. Analysis set and output completeness

The fixed-merge archive contained 40 model–seed records (five conditions and eight seeds), each evaluated on the same 455 held-out studies. Single retained outputs were available for image-only, text-only, and Gray-Square controls. GACR-B was post hoc and exploratory. Duplicate keys were resolved by the first-seen archive value; because a deterministic source-priority rule and hash manifest were not preserved, results remain tied to this snapshot. ECE and Brier score were recomputed from probability arrays in the companion pickle because the JSON summary omitted them. All confirmatory comparisons were paired at seed level; repeated test studies were not concatenated across seeds.

4.2. Predictive performance and calibration

Baseline CADQ achieved Macro-F1 0.9397 ± 0.0061 , accuracy 0.9607 ± 0.0032 , and Macro-AUROC 0.9974 ± 0.0004 (Table 1). LMH had the highest Macro-F1 (0.9435 ± 0.0093) but lower, more variable AUROC (0.9849 ± 0.0113). Noise-Consistency had the lowest confirmatory Macro-F1 (0.9346 ± 0.0075); Adaptive-CADQ was close to baseline. Exploratory GACR-B reached 0.9417 ± 0.0059 .

Table 1

Primary test-set performance across eight seeds (mean $\pm$ standard deviation [SD]). GACR-B was exploratory and post hoc.

Condition	Macro-F1	Accuracy	Macro-AUROC
Baseline CADQ	0.9397 ± 0.0061	0.9607 ± 0.0032	0.9974 ± 0.0004
LMH	0.9435 ± 0.0093	0.9596 ± 0.0070	0.9849 ± 0.0113
Noise-Consistency	0.9346 ± 0.0075	0.9560 ± 0.0039	0.9960 ± 0.0013
GACR-B (exploratory)	0.9417 ± 0.0059	0.9610 ± 0.0042	0.9972 ± 0.0005
Adaptive-CADQ	0.9388 ± 0.0072	0.9610 ± 0.0044	0.9972 ± 0.0008

Baseline per-class F1 was 0.9755 for Normal, 0.9624 for Pneumonia, and 0.8810 for Pneumothorax (Table 2). LMH improved Pneumothorax F1 to 0.9003 but reduced Pneumonia F1 to 0.9575. Noise-Consistency had the widest Pneumothorax variation; Adaptive-CADQ had the most stable Pneumonia F1.

Table 2

Calibration and per-class F1 across eight seeds (mean $\pm$ SD).

Condition	ECE	Brier	Normal F1	Pneumonia F1	Pneumothorax F1
Baseline CADQ	0.0237 ± 0.0048	0.0522 ± 0.0040	0.9755 ± 0.0032	0.9624 ± 0.0055	0.8810 ± 0.0174
LMH	0.0621 ± 0.0160	0.0685 ± 0.0213	0.9726 ± 0.0066	0.9575 ± 0.0077	0.9003 ± 0.0191
Noise-Consistency	0.0347 ± 0.0137	0.0655 ± 0.0130	0.9754 ± 0.0044	0.9530 ± 0.0058	0.8753 ± 0.0237
GACR-B (exploratory)	0.0245 ± 0.0023	0.0540 ± 0.0038	0.9741 ± 0.0043	0.9630 ± 0.0066	0.8880 ± 0.0167
Adaptive-CADQ	0.0241 ± 0.0070	0.0527 ± 0.0028	0.9776 ± 0.0054	0.9616 ± 0.0029	0.8772 ± 0.0175

No Macro-F1 variant contrast remained significant after Benjamini–Hochberg correction. The largest unadjusted signal was exploratory GACR-B ($\Delta = +0.0020$, Wilcoxon $p = 0.0547$, false-discovery rate (FDR) $p = 0.2188$). LMH

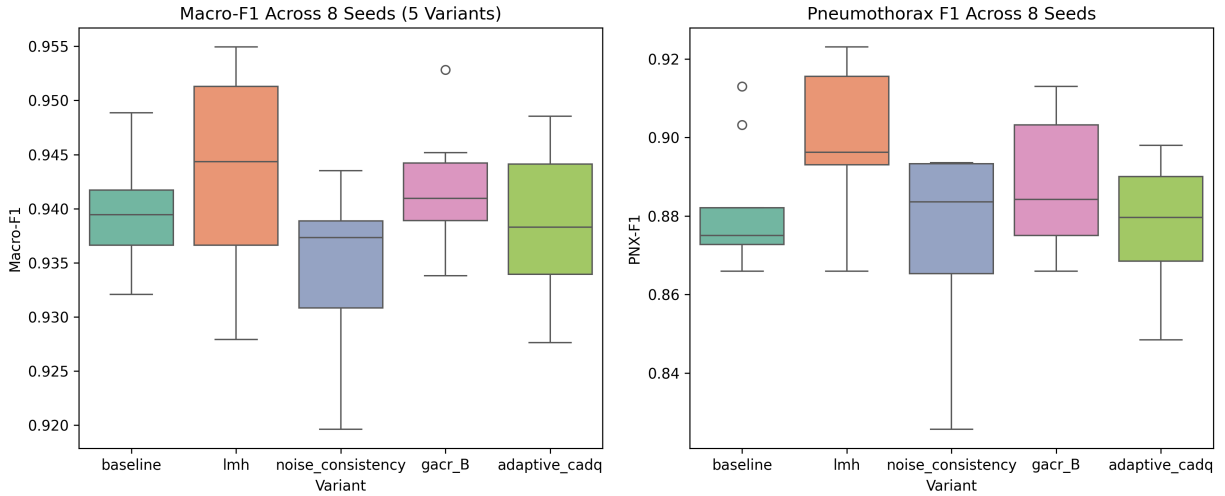


Figure 3: Macro-F1 and Pneumothorax F1 distributions across eight seeds.

reduced AUROC by 0.0126 (FDR $p = 0.000012$), and Noise-Consistency by 0.0014 (FDR $p = 0.0044$); Adaptive-CADQ did not differ from baseline (FDR $p = 0.8438$). LMH increased ECE by 0.0384 and Brier by 0.0163 (FDR $p = 0.0313$ and 0.0156). Noise-Consistency increased Brier by 0.0133 (FDR $p = 0.0156$), while its ECE change was inconclusive.

Table 3

Paired comparison p -values versus Baseline. Wilcoxon tests use eight seed-level Macro-F1 values; DeLong uses matched Pneumothorax ROC predictions (seed 42, the retained probability set); McNemar uses paired classification decisions on the same 455 studies. No predictions were pooled across seeds. Values in parentheses are Benjamini–Hochberg adjusted within each test family.

Condition	Wilcoxon Macro-F1 p (BH)	DeLong PNX AUROC p (BH)	McNemar p (BH)
LMH (PoE)	0.2500 (0.3333)	< 0.0001 (0.000012)	0.6936 (1.0000)
Noise-Consistency	0.1953 (0.3333)	0.0022 (0.0044)	0.0754 (0.3018)
GACR-B (exploratory)	0.0547 (0.2188)	0.0315 (0.0420)	1.0000 (1.0000)
Adaptive-CADQ	0.8438 (0.8438)	0.7304 (0.7304)	1.0000 (1.0000)

GACR-B and Adaptive-CADQ produced zero prediction discordance with Baseline across all 455 held-out studies (McNemar $p = 1.000$ for each). Thus, despite different loss functions and training dynamics, these variants yielded byte-identical decisions on the test set. This result is consistent with a text-dominated optimization landscape in which the report shortcut acts as a single attractor; it is not evidence that the image pathway is intrinsically non-functional in other cohorts.

Mean row-normalized baseline recalls were 97.78% (Normal), 95.54% (Pneumonia), and 89.36% (Pneumothorax). LMH improved Normal recall but reduced Pneumothorax recall; Noise-Consistency showed the reverse trade-off.

The baseline reliability error was 0.0237 ± 0.0048 and Brier score 0.0522 ± 0.0040 ; LMH was less calibrated. Reliability and ROC displays are descriptive because the same studies recur across seeds.

4.3. Controls and report-preserving ablation

The retained image-only control achieved Macro-F1 0.4611, accuracy 0.5670, and AUROC 0.6728; text-only achieved 0.8555 and 0.9820, respectively. Gray-Square retained Macro-F1 0.9381, accuracy 0.9604, and AUROC 0.9960, only 0.0015 below the baseline mean. These are single saved runs, and image-only checkpoint selection was imperfect, so the controls are diagnostic rather than modality rankings.

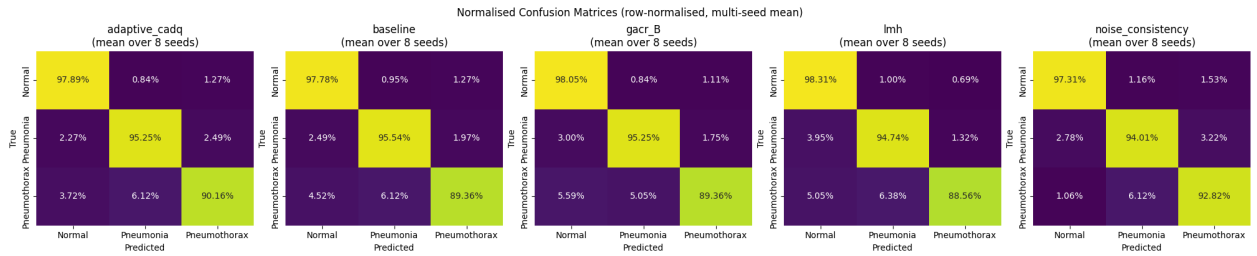


Figure 4: Row-normalized confusion matrices aggregated across the eight seeds. Each matrix is normalized by the true-class row; the displayed aggregate is descriptive because the same held-out studies recur across seeds.

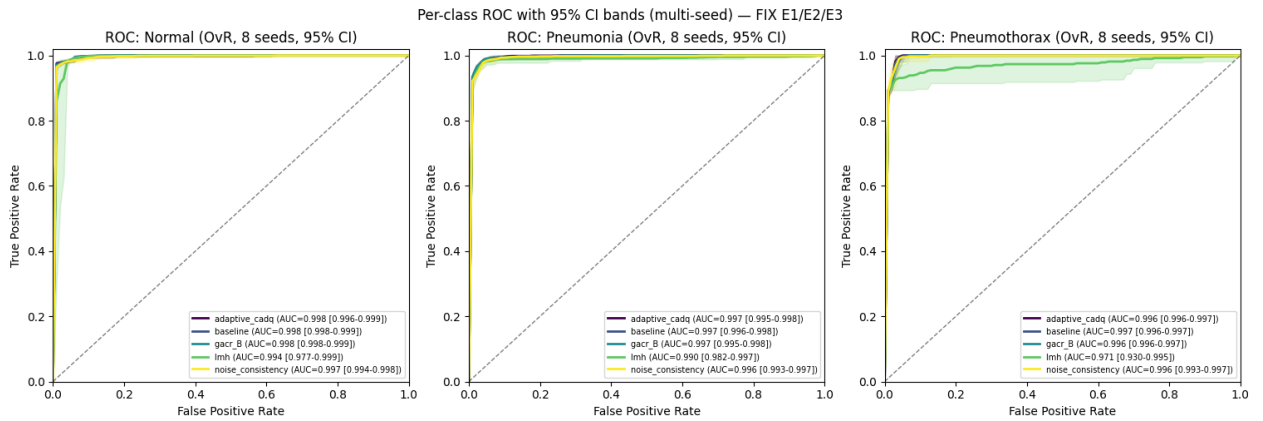


Figure 5: Per-class ROC curves with descriptive multi-seed 95% bands. Significant DeLong p -values for LMH and Noise-Consistency reflect AUROC degradation relative to Baseline, not improvement; all DeLong comparisons use matched seed-level predictions.

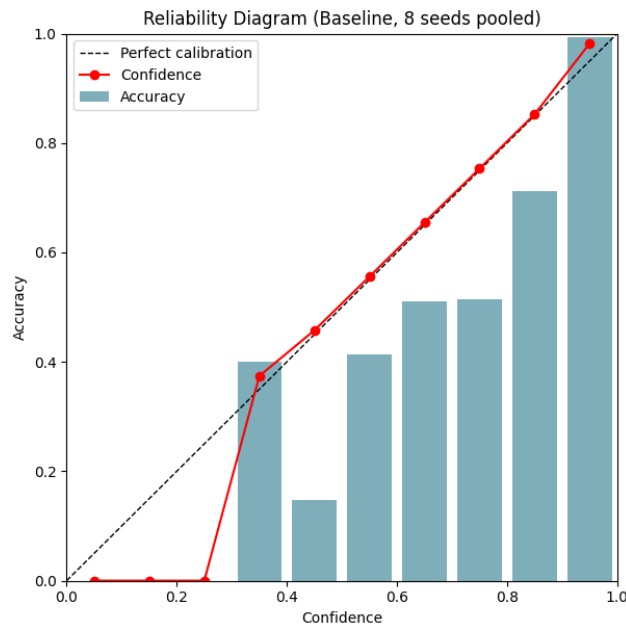


Figure 6: Descriptive baseline reliability diagram.

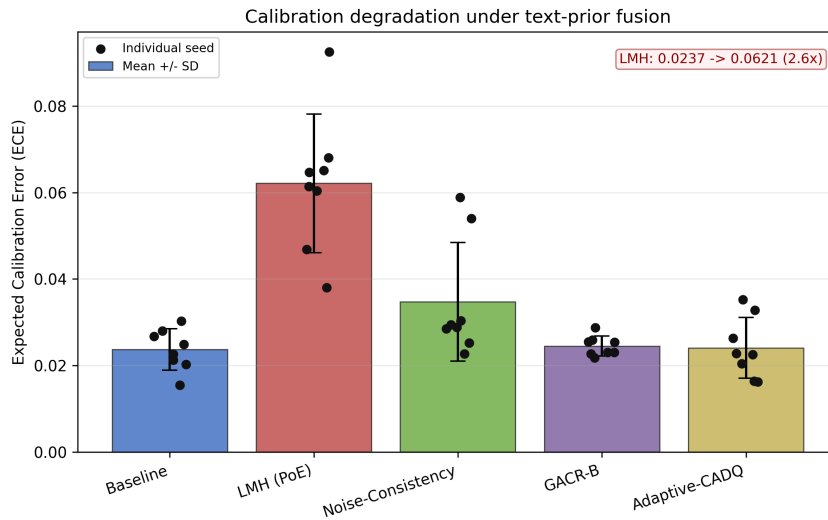


Figure 7: Expected calibration error (ECE) by condition with individual seed values. LMH increases ECE from 0.0237 to 0.0621 (2.6-fold); bars show mean $\pm$ SD across eight seeds.

Table 4

Retained control outputs on the 455-study test set (single runs).

Control	Macro-F1	Accuracy	Macro-AUROC	ECE	Brier
Image-only	0.4611	0.5670	0.6728	0.0993	0.6142
Text-only	0.8555	0.9077	0.9820	0.0203	0.1289
Gray-Square	0.9381	0.9604	0.9960	0.0369	0.0589

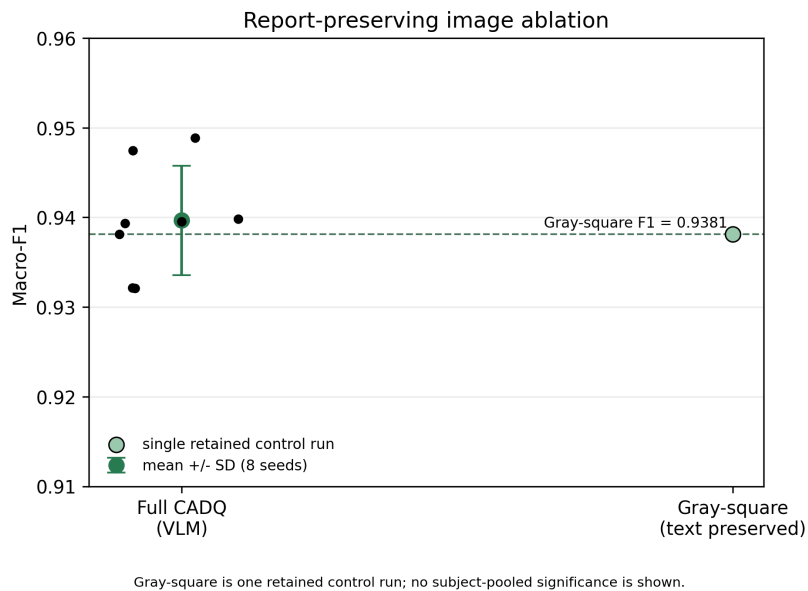


Figure 8: Report-preserving image ablation; Gray-Square is one retained run.

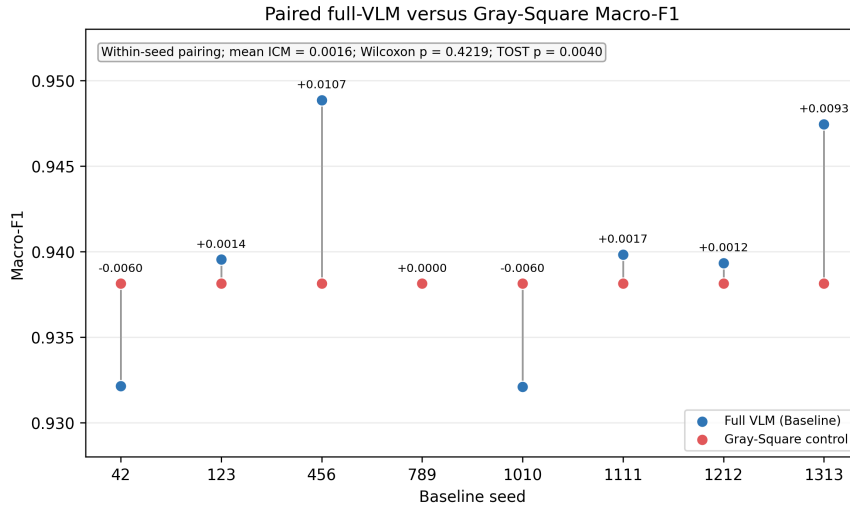


Figure 9: Paired Baseline VLM and Gray-Square Macro-F1 for the eight baseline seeds. Lines connect the two predictions for the same seed; the small paired differences make the near-zero ICM visible.

4.4. Image Contribution Metric

With Gray-Square as reference, baseline ICM was 0.0016 ± 0.0065 (median 0.0014). The bootstrap 95% confidence interval (CI) was $[-0.0026, 0.0061]$. The signed Wilcoxon test did not reject zero ($p = 0.4219$), while TOST supported equivalence within $[-0.01, 0.01]$ ($p = 0.0040$, Cohen's $d = 0.247$). LMH had ICM 0.0056 ± 0.0098 (TOST $p = 0.1223$); Noise-Consistency -0.0039 ± 0.0081 (TOST $p = 0.0345$); Adaptive-CADQ 0.0007 ± 0.0077 (TOST $p = 0.0054$); and exploratory GACR-B 0.0038 ± 0.0062 (TOST $p = 0.0122$). None demonstrated a robust increase in incremental image contribution.

Table 5

Image Contribution Metric (ICM) statistics by condition (eight seeds). Confidence intervals are percentile bootstrap intervals from 2,000 resamples; Wilcoxon tests are signed-rank tests against zero; TOST tests equivalence to zero within $[-0.01, 0.01]$.

Condition	Mean	SD	Median	95% CI	Wilcoxon p	TOST p	Cohen's d
Baseline	0.0016	0.0065	0.0014	$[-0.0026, 0.0061]$	0.4219	0.0040	0.247
LMH (PoE)	0.0056	0.0098	0.0066	$[-0.0011, 0.0118]$	0.1953	0.1223	0.566
Noise-Consistency	-0.0039	0.0081	-0.0009	$[-0.0097, 0.0009]$	0.3828	0.0345	-0.476
GACR-B (exploratory)	0.0038	0.0062	0.0030	$[-0.0002, 0.0081]$	0.1953	0.0122	0.607
Adaptive-CADQ	0.0007	0.0077	0.0002	$[-0.0043, 0.0057]$	0.8438	0.0054	0.089

4.5. Fusion gates and qualitative inspection

Extracted baseline image-gate means were 0.42624 (Normal), 0.49963 (Pneumonia), and 0.57336 (Pneumothorax), with overall mean 0.49975 versus 0.50000 at initialization. The largest class-wise drift was below 0.0004; the gate neither collapsed to 0.20 nor demonstrated learned image reliance. Three Pneumonia Grad-CAM examples (Fig. 12) are qualitative only and do not support localization or causal-grounding claims.

Overall, high performance persisted after image replacement, baseline ICM was near zero, and the gate remained near initialization. This pattern is consistent with report-linked shortcut potential in this cohort, not a universal claim that multimodal models ignore images.

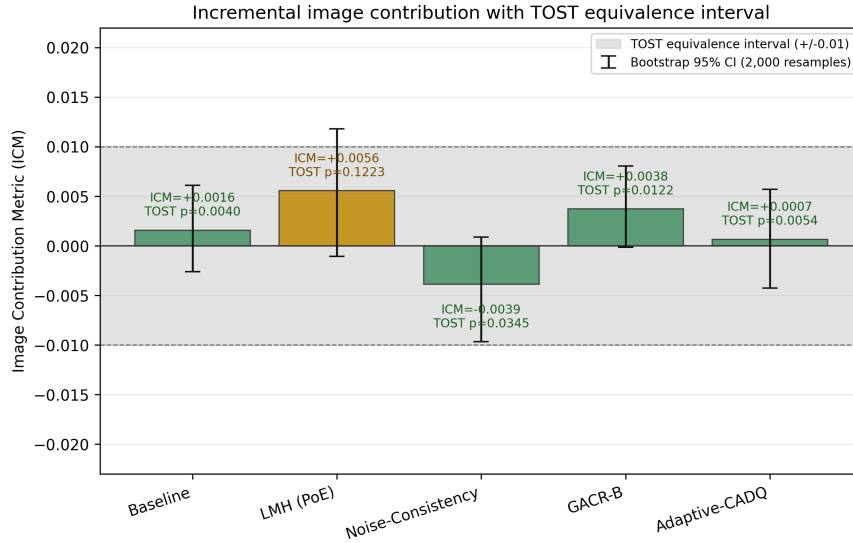


Figure 10: ICM by condition with percentile bootstrap 95% confidence intervals. The shaded band is the pre-specified TOST equivalence interval $[-0.01, 0.01]$; corrected Master statistics are annotated for each condition.

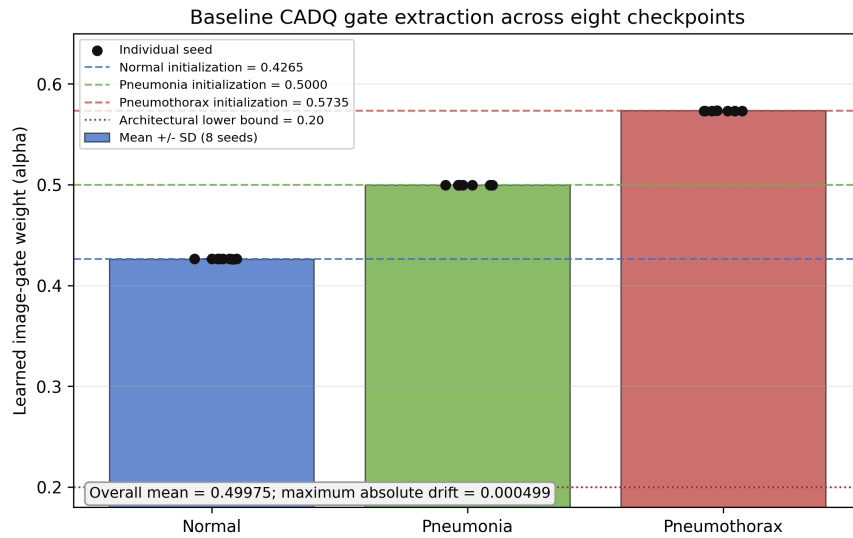


Figure 11: Baseline CADQ image-gate weights extracted from eight checkpoints. Bars show mean $\pm$ SD, dots show individual seeds, and dashed lines show the class-specific initialization values computed from the saved checkpoint logits.

Table 6

Per-seed CADQ gate extraction. Initialization values are computed from $u = (-0.5, 0, 0.5)$ and the bounded gate equation; maximum drift is the largest absolute class-wise deviation from initialization for that seed.

Seed	Normal α	Pneumonia α	Pneumothorax α	Max drift
42	0.426270	0.499512	0.573242	0.000488
123	0.426270	0.499512	0.573730	0.000488
456	0.426025	0.499756	0.573242	0.000499
789	0.426270	0.499756	0.573242	0.000255
1010	0.426270	0.499756	0.573242	0.000255
1111	0.426270	0.499512	0.573242	0.000488
1212	0.426270	0.499756	0.573730	0.000255
1313	0.426270	0.499512	0.573242	0.000488

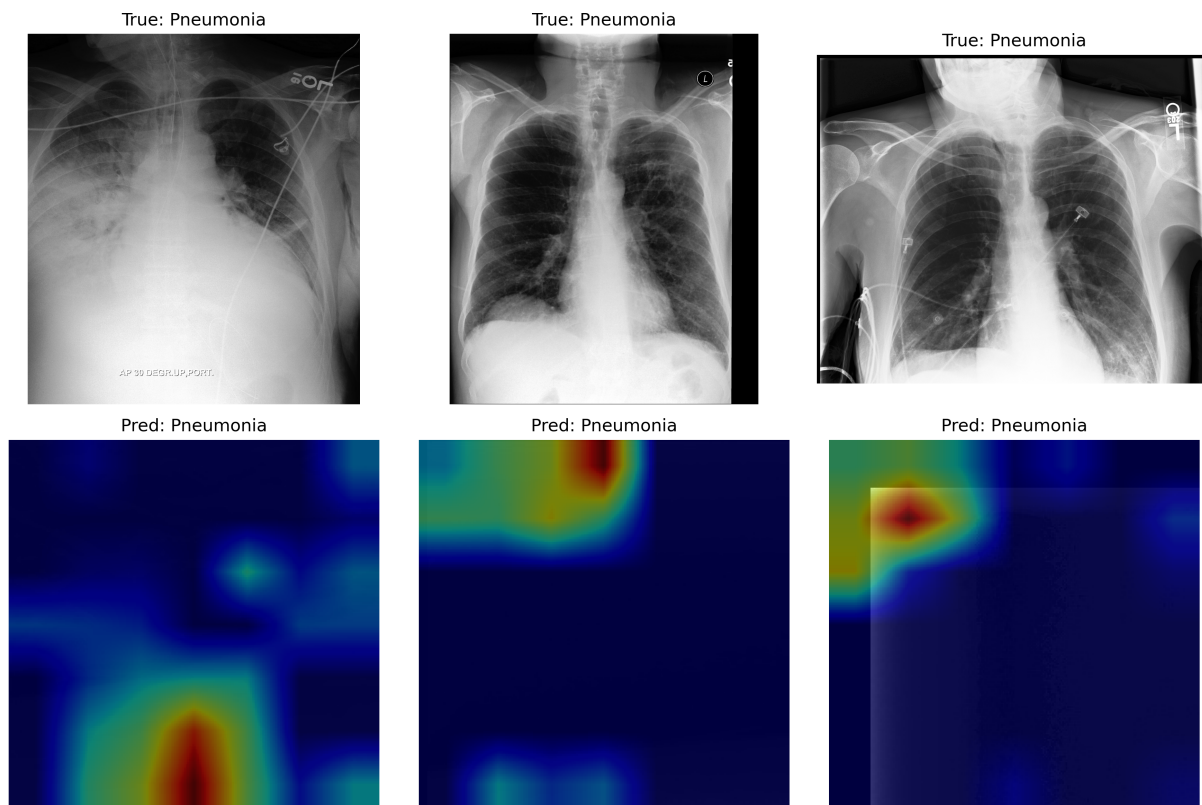


Figure 12: Qualitative Grad-CAM examples (method of Selvaraju et al. (Selvaraju et al., 2017)); no quantitative localization claim.

5. Discussion

5.1. Principal findings

The central result was a mismatch between high predictive performance and incremental image evidence. Gray-Square retained Macro-F1 0.9381 versus the baseline eight-seed mean of 0.9397 ± 0.0061 ; baseline ICM was 0.0016 ± 0.0065 , with a non-significant signed test and TOST equivalence to the operational interval $[-0.01, 0.01]$. Thus, replacing the radiograph changed little under this intervention. It does not imply that radiographs are universally unused.

The tested interventions did not provide a consistent remedy. LMH had the highest Macro-F1 but lower AUROC and substantially worse ECE and Brier score; Noise-Consistency also increased Brier score without increasing ICM. Adaptive-CADQ remained near baseline, and the small GACR-B increase was exploratory because it was added after the initial plan. Extracted image gates (0.42624, 0.49963, and 0.57336 by class) stayed near initialization rather than collapsing. This is a stagnation/null observation, not proof that the image branch was disconnected.

We refer to this pattern as modality inertia, or gate stagnation: the gate neither collapses nor learns because the report pathway already supplies a strong label-correlated signal, leaving insufficient gradient incentive for image reliance.

5.2. Label-report circularity and endpoint validity

MIMIC-CXR radiographs are paired with reports, CheXpert labels are generated from report language, and the model receives the report’s findings field (Johnson et al., 2019; Irvin et al., 2019). The task therefore does not isolate image-based disease recognition. Text-only Macro-F1/AUROC were 0.8555/ 0.9820, while Gray-Square reached 0.9381/0.9960. A constant image consequently preserved most of the multimodal score, consistent with report-linked

shortcut potential. The control does not identify the phrases used or show that labels are clinically wrong; it identifies a construct-validity risk in which the target and input share an information source.

This linkage explains why text-debiasing can appear incoherent. Suppressing a report shortcut may remove information rewarded by a report-derived label without creating an independent visual signal, yielding lower discrimination or calibration with no ICM gain. The null result should therefore not be attributed solely to a failed fusion module.

The GACR-B result is especially informative. Even the grounding-aware consistency intervention proposed by Sadanandan and Behzadan (Sadanandan and Behzadan, 2026) could not overcome the circularity: with $\lambda = 0.3$, its consistency penalty did not generate enough gradient to counter the dominant report shortcut. This is a limitation of the label–input relationship, not a claim that the proposed intervention is ineffective under independent labels.

5.3. Architectural and statistical interpretation

The attention block contains one projected image token and one projected text token. Its single-key attention cannot express word–region alignment, so its weights and Grad-CAM examples are not causal grounding evidence. High Macro-F1, near-zero ICM, and a non-collapsed gate can coexist when the image is redundant with text or receives little optimisation incentive. The archived negative-KL Noise-Consistency objective additionally rewards prediction divergence when minimised; a conventional consistency claim requires a positive-KL rerun.

The Gray-Square control was a single retained run reused for eight ICM values, and image-only checkpoint restoration was inconsistent. These asymmetries make the controls useful diagnostics but understate control uncertainty and prevent modality ranking. Confirmatory work should train matched controls at every seed and propagate their variation. Seed-level pairing is appropriate because the same 455 studies recur; pooled subject results are descriptive only.

5.4. Implications for evaluation and future study design

Modality contribution should be audited directly rather than inferred from a high multimodal score. Future studies should combine report-preserving image ablation with independent or image-derived labels, masked or temporally separated report fields, counterfactual text tests, deterministic source provenance, and external cohorts. ICM should be validated under these settings, not treated as a standalone grounding certificate. Reporting controls, label provenance, checkpoint rules, calibration, and seed-level uncertainty together is consistent with CLAIM and TRIPOD+AI transparency principles (Mongan et al., 2024; Collins et al., 2024).

For regulatory-facing evaluation, ICM and gate-stagnation checks should be treated as evidence-traceability tests rather than optional visualisations. A high aggregate score does not establish that the declared image input affected the decision, particularly when labels and reports share provenance. FDA good machine-learning-practice principles and its real-world-evidence guidance emphasise traceable data provenance, representative evaluation, uncertainty, and evidence that supports the intended use (U.S. Food and Drug Administration, 2025a,b). In that spirit, a submission for a multimodal medical device should report the report-preserving ablation, matched seed-level uncertainty, calibration, gate behaviour, and independent-label validation before making a visual-grounding claim.

The findings are not a clinical recommendation to ignore radiographs. They are limited to one report-linked cohort, task, preprocessing path, and checkpoint archive, with known normalisation, loss-sign, small-minority, post-hoc, and single-control limitations. The defensible conclusion is that these interventions did not establish robust incremental image evidence in this benchmark; corrected implementations, independent labels, matched controls, and external validation are needed before broader claims.

6. Limitations

6.1. Construct validity and cohort scope

The principal limitation is label provenance. CheXpert targets and the model’s findings input both derive from the MIMIC-CXR reports (Johnson et al., 2019; Irvin et al., 2019); the controls expose this overlap but cannot identify the phrases used or establish label error. Independent image-derived adjudication, report masking, or a temporally separated target is needed to claim visual diagnosis. The study also covers only three classes, 2,953 usable studies, and 47 test Pneumothorax cases. Class caps, uncertain-label handling, prevalence, and patient-level split choices limit extrapolation.

6.2. Implementation and architecture

The archived image mapping to $[-1, 1]$ may not match the native TorchXRayVision preprocessing expected by the selected weights (Cohen et al., 2022). The negative-KL Noise-Consistency loss rewards divergence in a minimised objective, so it cannot support a conventional consistency claim until rerun with positive KL. One image token and one text token make the eight-head attention feature-level rather than word-region alignment; gates and Grad-CAM are therefore diagnostic, not causal.

6.3. Controls and uncertainty

Gray-Square was one retained run reused for eight ICM values, while image-only checkpoint restoration was inconsistent. Controls are consequently diagnostic, not modality rankings, and a matched multi-seed design is required. Seed-level inference leaves only $n = 8$ replications; the operational TOST interval $[-0.01, 0.01]$ is a project-defined margin, not a clinical or regulatory threshold (Lakens, 2017).

6.4. Provenance and generalisability

The fixed archive retained first-seen duplicate merge values without a preserved source-priority rule or hash manifest. Adaptive-CADQ's eight-seed extension and GACR-B were post hoc, and probability arrays had to be recovered from a companion pickle. Future releases should lock notebooks, checkpoints, configurations, probabilities, source hashes, and deterministic merge rules. No external cohort, prospective workflow, reader study, calibration transfer, or outcome analysis was performed. Reports, acquisition protocols, prevalence, and label processes may differ elsewhere. Correct preprocessing and loss, matched controls, independent labels, counterfactual report tests, and external validation are required before general or clinical claims.

7. Conclusion

On this MIMIC-CXR-derived task, replacing radiographs with a constant Gray-Square input preserved nearly the baseline Macro-F1, while the eight-seed ICM was near zero and equivalent to the operational null interval. Tested fusion and text-debiasing interventions did not establish a larger incremental image contribution, and the baseline gate remained near initialization.

The result is a benchmark warning, not proof that radiographs are universally unused. CheXpert labels and the model's findings input share report provenance, allowing report-linked shortcuts. Corrected preprocessing and loss implementations, matched controls, independent labels, deterministic provenance, counterfactual text tests, and external validation are needed before broader or clinical claims. ICM is best treated as a diagnostic audit measure, not a standalone grounding certificate.

8. Reporting statements

Ethics approval and consent

MIMIC-CXR is de-identified and was accessed through official PhysioNet credentialing and its Data Use Agreement. No participants, direct identifiers, or identifying images were involved; consent was not applicable. The source resource was released under its institutional review-board approvals; this secondary analysis involved no new recruitment, intervention, or contact with participants. Any local institutional exemption or waiver identifier can be added here if the target institution requires one.

Data availability

Raw MIMIC-CXR images and reports remain under PhysioNet access controls and are not redistributed. Notebooks, scripts, manifests, configurations, figures, and non-identifying summaries are available at <https://github.com/i-mTanvir/MediVision>. The release includes environment information, source history, and the documented merge rule.

Code availability

Reproducible notebooks, LaTeX source, figure scripts, and environment records are available at <https://github.com/i-mTanvir/MediVision>. A permanent DOI will be added if an archival release is deposited. Credentials, raw reports, and identifiers are excluded.

Funding

This research received no specific public, commercial, or not-for-profit grant and was self-funded by the authors. No external sponsor influenced design, data access, analysis, interpretation, writing, or the decision to submit.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced this work.

Author contributions

Nafis Al Rahman and Tanvir Mahmud contributed equally to conceptualization, methodology, investigation, software, analysis, visualization, and the original draft. Both validated, interpreted, revised, and approved the manuscript.

Acknowledgements

The authors thank PhysioNet and the MIMIC-CXR data custodians for maintaining the de-identified resource and controlled-access infrastructure.

Generative-AI disclosure

Generative-AI tools were not used to generate study data, execute analyses, create scientific results, or make study decisions. Any language or formatting assistance used during manuscript preparation was reviewed and approved by both authors, who accept responsibility for the final content.

Supplementary materials

Table S1. Training hyperparameters

Setting	Value
Image encoder	TorchXRayVision DenseNet-121, res224 MIMIC weights
Text encoder	Bio_ClinicalBERT; first four transformer layers frozen
Optimizer	AdamW; weight decay 10^{-4}
Learning rates	10^{-5} image/text; 10^{-4} heads; 5×10^{-5} static gates
Schedule	OneCycleLR with 10% warm-up; 10 epochs
Batching	Physical batch 8; four accumulation steps; effective batch 32
Objective	Focal cross-entropy ($\gamma = 1.5$), label smoothing 0.05, effective-number $\beta = 0.99$
Precision	FP16 mixed precision with gradient scaling
Gate initialisation	$u = (-0.5, 0, 0.5)$; $\alpha = (0.426524, 0.500000, 0.573476)$
Augmentation	Training crop/flip/rotation/translation/jitter; class-aware minority transform
Seeds	42, 123, 456, 789, 1010, 1111, 1212, 1313
Hardware	Kaggle NVIDIA T4 (two-GPU environment)

Table S2. CLAIM 2024 checklist summary

Status	Count	Interpretation
Yes	13	Reported or directly auditable in the manuscript/archive
Partial	10	Partly reported or limited by inherited archive constraints
No	4	Not available in the inherited experiment

Table S3. TRIPOD+AI checklist summary

Status	Count	Interpretation
Yes	9	Reported or directly auditable
Partial	18	Partly reported or limited by inherited archive constraints
No	14	Not available in the inherited experiment

Table S4. Provenance manifest

Condition	Source notebook	Seeds	Retained outputs	Known defect or caveat
Baseline CADQ	medvision-q1-audit-final-FIXED-MERGE-v2.ipynb	8	Predictions, probabilities, gate	Duplicate merge snapshot
LMH (PoE)	Same fixed-merge notebook	8	Predictions, probabilities	Lower AUROC and calibration
Noise-Consistency	Same fixed-merge notebook	8	Predictions, probabilities	Archived negative-KL sign
GACR-B	Same fixed-merge notebook	8 post hoc	Predictions, probabilities	Added after initial plan
Adaptive-CADQ	Same fixed-merge notebook	8 (post hoc extension)	Predictions, probabilities	Seed extension post hoc; gate extraction absent
Gray-Square	Audit archive	Single retained run	Predictions, probabilities	Reused control for paired ICM

References

- Agrawal, A., Batra, D., Parikh, D., Kembhavi, A., 2018. Don't just assume; look and answer: Overcoming priors for visual question answering. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.
- Alsentzer, E., Murphy, J., Boag, W., Weng, W.H., Jindi, D., Naumann, T., McDermott, M., 2019. Publicly available clinical bert embeddings, in: *Proceedings of the 2nd Clinical Natural Language Processing Workshop, Association for Computational Linguistics*. pp. 72–78. doi:10.18653/v1/W19-1909.
- Arevalo, J., Solorio, T., Montes-y Gómez, M., González, F.A., 2017. Gated multimodal units for information fusion. *arXiv preprint arXiv:1702.01992* URL: <https://arxiv.org/abs/1702.01992>.
- Banerjee, I., Bhattacharjee, K., Burns, J.L., Trivedi, H., Purkayastha, S., Seyyed-Kalantari, L., Patel, B.N., Rakesh, S., Gichoya, J., 2023. Shortcuts causing bias in radiology artificial intelligence: Causes, evaluation, and mitigation. *Journal of the American College of Radiology*.
- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57, 289–300. doi:10.1111/j.2517-6161.1995.tb02031.x.
- Boecking, B., Usuyama, N., Bannur, S., Castro, D.C., Schwaighofer, A., Hyland, S., Wetscherek, M., Naumann, T., Nori, A., Alvarez-Valle, J., Poon, H., Oktay, O., 2022. Making the most of text semantics to improve biomedical vision-language processing. *arXiv preprint arXiv:2204.09817*.
- Cadene, R., Dancette, C., Cord, M., Parikh, D., et al., 2019. Rubi: Reducing unimodal biases for visual question answering. *arXiv preprint arXiv:1906.10169*.
- Chiang, L.F., 2026. An interpretable chest x-ray classification framework using prototype memory and counterfactual consistency. *Cureus* 18, e103134.
- Clark, C., Yatskar, M., Zettlemoyer, L., 2019. Don't take the easy way out: Ensemble based methods for avoiding known dataset biases. *arXiv preprint arXiv:1909.03683*.
- Cohen, J.P., Viviano, J.D., Bertin, P., Morrison, P., Torabian, P., Guarrera, M., Lungren, M.P., Chaudhari, A., Brooks, R., Hashir, M., Bertrand, H., 2022. Torchxrayvision: A library of chest x-ray datasets and models, in: *Proceedings of the 5th International Conference on Medical Imaging with Deep Learning, PMLR*. pp. 231–249. URL: <https://proceedings.mlr.press/v172/cohen22a.html>.
- Collins, G.S., Moons, K.G.M., Dhiman, P., et al., 2024. Tripod+ai statement: Updated guidance for reporting prediction models that use regression or machine learning methods. *The BMJ* 385, e078378. doi:10.1136/bmj-2023-078378.
- DeLong, E.R., DeLong, D.M., Clarke-Pearson, D.L., 1988. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics* 44, 837–845. doi:10.2307/2531595.
- Devlin, J., Chang, M.W., Lee, K., Toutanova, K., 2019. Bert: Pre-training of deep bidirectional transformers for language understanding, in: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Association for Computational Linguistics*. pp. 4171–4186. doi:10.18653/v1/N19-1423.
- Geirhos, R., et al., 2020. Shortcut learning in deep neural networks. *Nature Machine Intelligence*.
- Huang, G., Liu, Z., van der Maaten, L., Weinberger, K.Q., 2017. Densely connected convolutional networks, in: *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2261–2269. doi:10.1109/CVPR.2017.243.
- Huang, S.C., Shen, S., et al., 2021. Gloria: A multimodal global-local representation learning framework for label-efficient medical image recognition. *Proceedings of the IEEE/CVF International Conference on Computer Vision*.
- Irvine, J., Rajpurkar, P., Ko, M., Yu, Y., Ciurea-Ilcus, S., Chute, C., Marklund, H., Haghighi, B., Ball, R.L., Shpankaya, K.S., Seekins, J., Mong, D.A., Halabi, S.S., Sandberg, J.K., Jones, R., Larson, D.B., Langlotz, C.P., Patel, B.N., Lungren, M.P., Ng, A.Y., 2019. Chexpert: A large chest radiograph dataset with uncertainty labels and expert comparison, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 590–597. doi:10.1609/AAAI.V33I01.3301590.
- Jin, Q., et al., 2024. Hidden flaws behind expert-level accuracy of multimodal gpt-4 vision in medicine. *npj Digital Medicine*.
- Johnson, A.E.W., Pollard, T.J., Berkowitz, S.J., Greenbaum, N.R., Lungren, M.P., Deng, C.Y., Mark, R.G., Horng, S., 2019. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific Data* 6, 317. doi:10.1038/s41597-019-0322-0.
- Lakens, D., 2017. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science* 8, 355–362. doi:10.1177/1948550617697177.
- Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollár, P., 2017. Focal loss for dense object detection, in: *2017 IEEE International Conference on Computer Vision (ICCV)*, pp. 2999–3007. doi:10.1109/ICCV.2017.324.
- Loshchilov, I., Hutter, F., 2019. Decoupled weight decay regularization, in: *7th International Conference on Learning Representations*. URL: <https://openreview.net/forum?id=Bkg6RiCqY7>.
- Lotfinia, M., Ziegelmayer, S., Adams, L., Truhn, D., Maier, A., Tayebi Arasteh, S., 2026. Vision-language models for chest radiography do not always need the image. *arXiv preprint arXiv:2606.17710*.
- McNemar, Q., 1947. Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika* 12, 153–157. doi:10.1007/BF02295996.
- Mongan, J., Moy, L., Erickson, B.J., et al., 2024. Checklist for artificial intelligence in medical imaging (claim): A guide for authors and reviewers. *Radiology: Artificial Intelligence* 6, e230120. doi:10.1148/ryai.230120.
- Myronenko, A., Yang, D., Turkbey, B., Aboian, M., Azamat, S., Akcicek, E., Yin, H., Molchanov, P., Edgar, M., He, Y., Guo, P., Tang, Y., Xu, D., 2025. Reasoning visual language model for chest x-ray analysis. *arXiv preprint arXiv:2510.23968*.
- Sabottke, C., Lee, J., Chiang, A., Spieler, B., Mushtaq, R., 2024. Text report analysis to identify opportunities for optimizing target selection for chest radiograph artificial intelligence models. *Journal of Imaging Informatics in Medicine* 37, 402–411.
- Sadanandan, B., Behzadan, V., 2026. Consistent but dangerous: Per-sample safety classification reveals false reliability in medical vision-language models. *arXiv preprint arXiv:2603.20985*.
- Santomartino, S.M., Zech, J.R., Hall, K., Jeudy, J., Parekh, V., Yi, P.H., 2024. Evaluating the performance and bias of natural language processing tools in labeling chest radiograph reports. *Radiology*.

- Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization, in: 2017 IEEE International Conference on Computer Vision (ICCV), pp. 618–626. doi:10.1109/ICCV.2017.74.
- Smith, L.N., Topin, N., 2019. Super-convergence: Very fast training of neural networks using large learning rates. *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications* 11006, 1100612. doi:10.1117/12.2520589.
- U.S. Food and Drug Administration, 2025a. Good machine learning practice for medical device development: Guiding principles. FDA guidance. URL: <https://www.fda.gov/medical-devices/software-medical-device-samd/good-machine-learning-practice-medical-device-development-guiding-principles>. accessed 2026-08-31.
- U.S. Food and Drug Administration, 2025b. Use of real-world evidence to support regulatory decision-making for medical devices. FDA guidance. URL: <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/use-real-world-evidence-support-regulatory-decision-making-medical-devices>. accessed 2026-08-31.
- Wang, A., Zhang, J., Oberst, M., 2026. Revisiting performance claims for chest x-ray models using clinical context. *Proceedings of Machine Learning Research* 333.
- Zhang, Y., Jiang, H., Miura, Y., Manning, C.D., Langlotz, C.P., 2022. Contrastive learning of medical visual representations from paired images and text. *Proceedings of Machine Learning Research* 182.